Annexure 2

DATA COMMUNICATION AND COMPUTER NETWORKS

Course Code : CSE 601 **Credit Units: 03**
Total Hours: 45

Course Objectives:

- The objective of the course is to provide knowledge of various multiplexing techniques and communication medias for transmission of data.
- Learn how computer network hardware and software operate.
- Investigate the fundamental issues driving network design Learn about dominant network technologies ,routing techniques ,error detection and correction techniques, various protocols and their working.

Course Contents:

Module I: Introduction: (7 Hours)
Introduction to Data Communication, Networks-protocols, advantages, disadvantages & applications, Line Configuration, topology, Transmission mode, Classification of networks. . OSI & TCP/IP reference models, with functionality and design issues of all layers presented in the models, LAN, MAN, WAN.
Parallel & Serial Transmissions, Analog & Digital Signals, Periodic & Aperiodic Signals, Modulation-Amplitude Modulation, Frequency Modulation, Phase Modulation, Pulse Amplitude Modulation, Pulse Code Modulation.

Module II: Physical Layer: (7 Hours)
TDM, FDM, WDM; Circuit switching time division & space division switch. Transmission Media-Twisted Pair Cable, Coaxial Cable, Fiber-Optics Cable, Radio frequency Allocation, Terrestrial Microwave, Infrared rays, Satellite Communication, Cellular Telephony.

Module III: Data Link Layer: (9 Hours)
Framing, Line Discipline, Types of Errors, Error Detection & Correction (VRC, LRC, CRC, Checksum, Hamming Code), Flow Control, Error Control, CSMA/CD, Project 802, IEEE Standards-802.3, Token Bus (802.4), Token Ring (802.5).

Module IV: Network Layer: (9 Hours)
Internetworking & devices: Repeaters, Hubs, Bridges, Switches, Router, Gateway; Internet address, classful address, subnetting; Static vs. dynamic routing , Routing algorithms: shortest path algorithm, flooding, distance vector routing, link state routing; Protocols: ARP,RARP, IP, ICMP and IPV6

Module V: Transport Layer: (7 Hours)
Process to process delivery; UDP; TCP; Congestion control algorithm: Leaky bucket algorithm, Token bucket algorithm, choke packets; Quality of service: techniques to improve Qos.

Module VI: Application Layer: (6 Hours)
DNS; SMTP, SNMP, FTP, HTTP & WWW; Security: Cryptography, user authentication, security protocols in internet, Firewalls.

Advanced topics
ISDN services & ATM; DSL technology, Wireless LAN: IEEE 802.11; Bluetooth, VLAN's, Cellular telephony & Satellite network.

Course Outcomes:

- Show clear understanding of the basic concepts of data communications including the key aspects of networking and their interrelationship, packet switching, circuit switching and cell switching as internal and external operations, physical structures, types, models, and internetworking.
- Demonstrate the ability to unambiguously explain networking as it relates to the connection of computers, media, and devices (routing).
- Able to intelligently compare and contrast local area networks and wide area networks in terms of characteristics and functionalities. Able to identify limitations of typical communication systems.
- Able to differentiate among and discuss the four levels of addresses (physical, logical, port, and specific used by the Internet TCP/IP protocols.